STATISTICS & MACHINE LEARNING · LESSON 14 OF 28

Classification Metrics: Confusion Matrix, Precision, Recall, ROC, PR and Business Value

Measure classifiers properly. The confusion matrix, accuracy and why it misleads, precision, recall, specificity, F1 and F-beta, ROC curves and AUC, precision-recall curves, log loss and calibration, lift and gains charts, choosing thresholds by cost or capacity, and multiclass averaging, using the telecom churn model.

ML class 7 | scikit-learn 1.7+ | Intermediate | 110 min

By the end of this lesson you can

- Build and read a confusion matrix and compute accuracy, precision, recall, specificity and F1
- Plot and interpret ROC and precision-recall curves and their AUCs
- Check probability calibration and log loss
- Choose thresholds by cost, capacity (top-k) or target recall, and present gains/lift
- Use macro, weighted and micro averages for multiclass problems

Datasets used

`telecom_churn.csv`

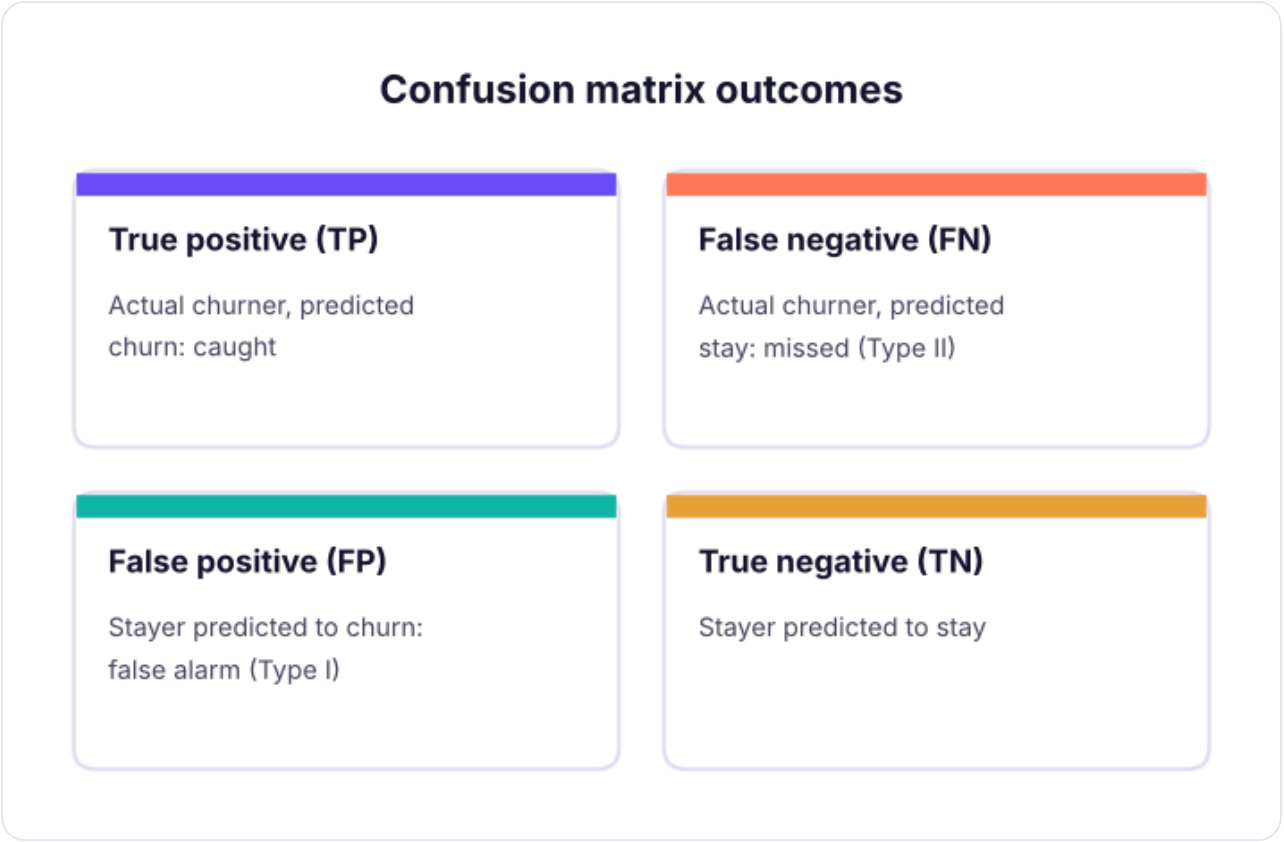
The confusion matrix

Python

```
import pandas as pd, numpy as np
from sklearn.model_selection import train_test_split
from sklearn.compose import ColumnTransformer
from sklearn.preprocessing import StandardScaler, OneHotEncoder
from sklearn.pipeline import Pipeline
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import confusion_matrix, accuracy_score, precision_score,
recall_score, f1_score, fbeta_score, classification_report
tc = pd.read_csv("telecom_churn.csv")
X = tc.drop(columns=["customer_id", "churn", "total_charges"]); y = (tc.churn ==
"Yes").astype(int)
Xtr, Xte, ytr, yte = train_test_split(X, y, test_size=0.3, stratify=y, random_state=1)
pre = ColumnTransformer([("n", StandardScaler(), ["senior_citizen", "tenure_months",
"monthly_charges", "support_calls"]), ("c", OneHotEncoder(), ["contract",
"internet_service", "payment_method"])])
model = Pipeline([("pre", pre), ("lr", LogisticRegression(max_iter=1000))]).fit(Xtr, ytr)
proba = model.predict_proba(Xte)[:, 1]; pred = (proba ⩾ 0.5).astype(int)
tn, fp, fn, tp = confusion_matrix(yte, pred).ravel()
print(pd.DataFrame([[tn, fp], [fn, tp]], index=["actual: stay", "actual: churn"], columns=
["pred: stay", "pred: churn"]))
```

OUTPUT

```
               pred: stay  pred: churn
actual: stay          611           57
actual: churn         125          107
```



The four outcomes of a binary classifier.

Metrics from the matrix

Metric	Formula	Question it answers
Accuracy	(TP + TN) / all	How often is the model right overall?
Precision	TP / (TP + FP)	Of those we flagged, how many were churners?
Recall (sensitivity, TPR)	TP / (TP + FN)	Of all churners, how many did we catch?
Specificity (TNR)	TN / (TN + FP)	Of all stayers, how many did we leave alone?
F1	2·P·R / (P + R)	Balance of precision and recall
F-beta	weights recall β times as much	β = 2 when misses are costlier

Python

```python
print(f"accuracy {accuracy_score(yte, pred):.3f} | precision {precision_score(yte, pred):.3f} | recall {recall_score(yte, pred):.3f} | specificity {tn / (tn + fp):.3f}")
print(f"F1 {f1_score(yte, pred):.3f} | F2 {fbeta_score(yte, pred, beta=2):.3f} | baseline accuracy (always 'stay') {1 - yte.mean():.3f}")
print(classification_report(yte, pred, target_names=["stay", "churn"], digits=3))
```

OUTPUT

```
accuracy 0.798 | precision 0.652 | recall 0.461 | specificity 0.915
F1 0.540 | F2 0.490 | baseline accuracy (always 'stay') 0.742
              precision    recall  f1-score   support

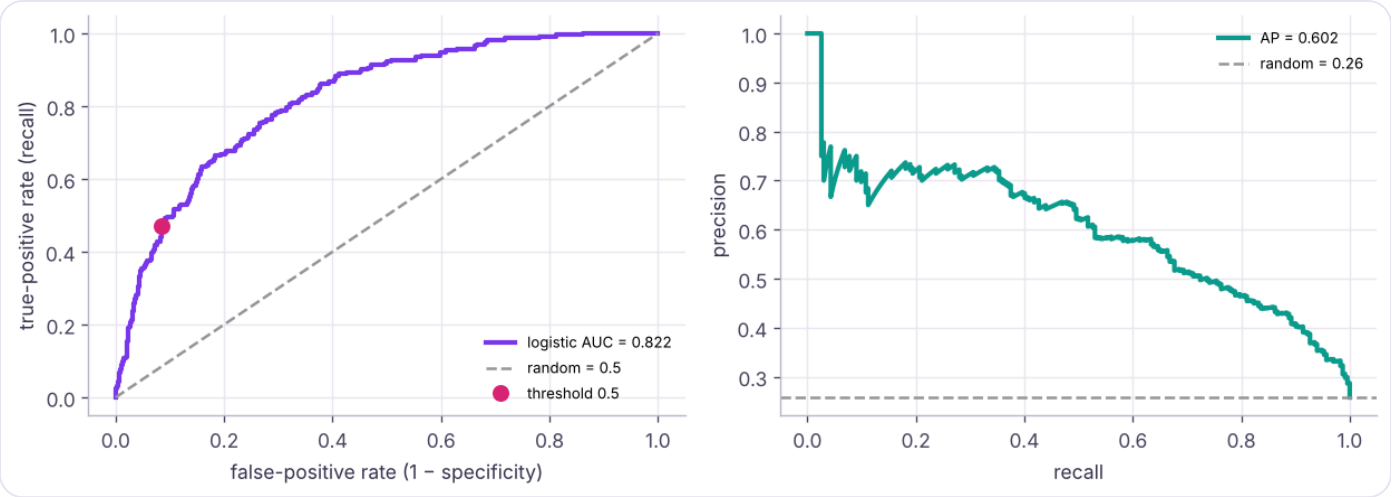
        stay      0.830     0.915     0.870       668
       churn      0.652     0.461     0.540       232

    accuracy                          0.798       900
   macro avg      0.741     0.688     0.705       900
weighted avg      0.784     0.798     0.785       900
```

Accuracy barely beats the "nobody churns" baseline, while recall shows the model misses more than half of the churners at the default 0.5 threshold.

Which metric matters?	Example
Recall	Cancer screening, fraud, churn (missing a positive is costly)
Precision	Spam filter (do not hide real mail), expensive interventions
Balance (F1)	Both errors matter similarly
Specificity	Avoiding false alarms for the majority

Threshold-free view: ROC and PR curves



Left: the ROC curve plots recall (TPR) against the false-positive rate at every threshold. Right: the precision-recall curve focuses on the positive class, more informative when positives are rare.

Python

```
print(f"ROC-AUC {roc_auc_score(yte, proba):.3f} | average precision (PR-AUC)
{average_precision_score(yte, proba):.3f} | churn base rate {yte.mean():.3f}")
```

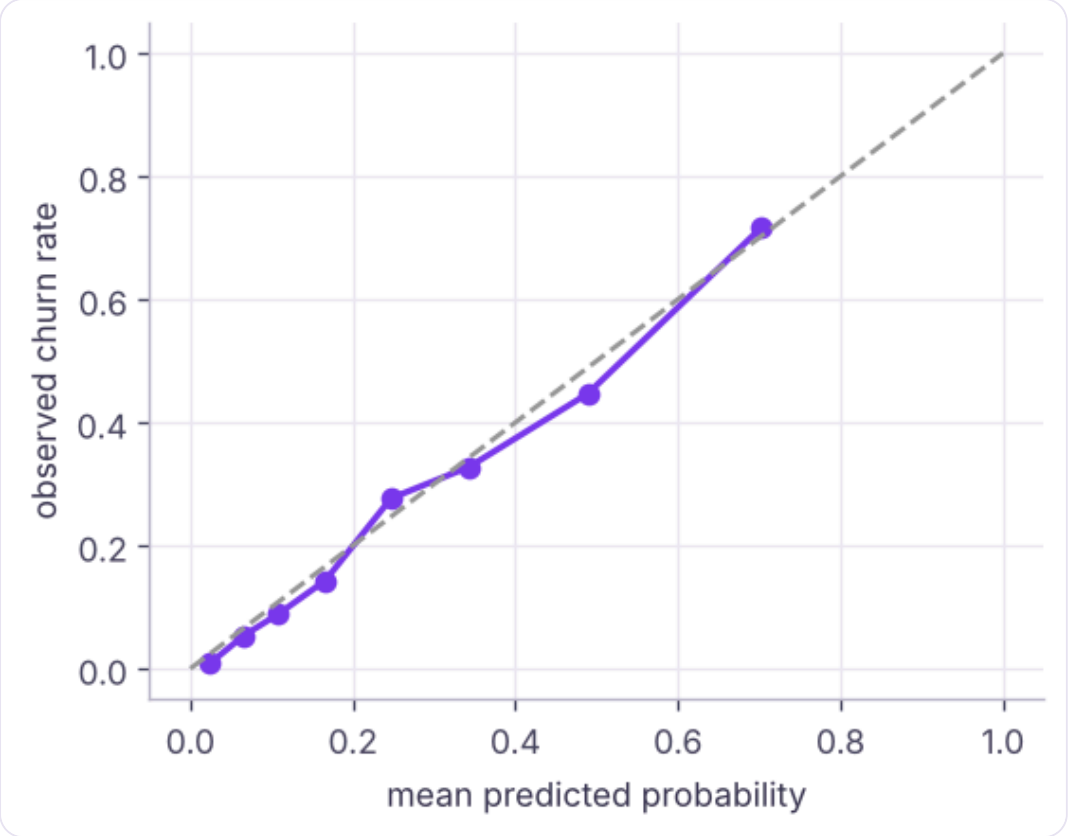
OUTPUT

```
ROC-AUC 0.822 | average precision (PR-AUC) 0.602 | churn base rate 0.258
```

ROC-AUC is the probability that a randomly chosen churner gets a higher score than a randomly chosen stayer (0.5 = random, 1 = perfect). It does not depend on the threshold or on class balance. **Average precision** summarises the PR curve; its random baseline equals the positive rate, so it reveals weakness on rare classes that ROC-AUC can hide.

Probability quality: log loss and calibration

If the model says 30%, do about 30% of those customers churn?



Calibration curve: points near the diagonal mean predicted probabilities can be trusted as probabilities.

Python

```
from sklearn.metrics import log_loss, brier_score_loss
print(f"log loss {log_loss(yte, proba):.3f} (always predicting the base rate:
{log_loss(yte, np.full(len(yte), ytr.mean())):.3f}) | Brier score {brier_score_loss(yte,
proba):.3f}")
```

OUTPUT

```
log loss 0.436 (always predicting the base rate: 0.571) | Brier score 0.142
```

Calibration matters when probabilities feed decisions (expected loss = P(default) × amount). Tree ensembles and SVMs are often poorly calibrated; fix with `CalibratedClassifierCV` (isotonic or sigmoid).

Choosing the threshold

By cost

Python

```
value_saved, offer_cost = 3000, 500          # a retained churner is worth ₹3,000; each
offer costs ₹500
rows = []
for t in np.round(np.arange(0.1, 0.81, 0.05), 2):
    p = proba ⩾ t; tp_ = (p & (yte == 1)).sum(); n_off = p.sum()
    rows.append({"threshold": t, "offers": n_off, "churners_reached": tp_, "precision":
round(tp_ / max(n_off, 1), 3), "recall": round(tp_ / yte.sum(), 3), "profit_inr": int(tp_ *
value_saved * 0.4 - n_off * offer_cost)})   # 40% of reached churners accept
res = pd.DataFrame(rows); print(res.to_string(index=False))
print("most profitable threshold:", res.loc[res.profit_inr.idxmax(), "threshold"])
```

OUTPUT

```
 threshold  offers  churners_reached  precision  recall  profit_inr
      0.10     637               221      0.347   0.953      -53300
      0.15     532               212      0.398   0.914      -11600
      0.20     448               197      0.440   0.849       12400
      0.25     394               184      0.467   0.793       23800
      0.30     326               165      0.506   0.711       35000
      0.35     272               151      0.555   0.651       45200
      0.40     226               132      0.584   0.569       45400
      0.45     191               119      0.623   0.513       47300
      0.50     164               107      0.652   0.461       46400
      0.55     130                87      0.669   0.375       39400
      0.60     101                72      0.713   0.310       35900
      0.65      77                55      0.714   0.237       27500
      0.70      54                39      0.722   0.168       19800
      0.75      35                25      0.714   0.108       12500
      0.80      20                15      0.750   0.065        8000
most profitable threshold: 0.45
```

By capacity: top-k, gains and lift

The retention team can call only 10% of customers per month. Rank by probability and take the top 10%.

Python

```
order = np.argsort(-proba); ys = yte.to_numpy()[order]
deciles = np.array_split(ys, 10)
gains = pd.DataFrame({"decile": range(1, 11), "churn_rate": [d.mean() for d in deciles]})
gains["cum_%_of_churners"] = (100 * np.cumsum([d.sum() for d in deciles]) /
ys.sum()).round(1)
gains["lift"] = (gains.churn_rate / ys.mean()).round(2); gains["churn_rate"] =
gains.churn_rate.round(3)
print(gains.to_string(index=False))
```

OUTPUT

decile	churn_rate	cum_%_of_churners	lift
1	0.722	28.0	2.80
2	0.556	49.6	2.16
3	0.400	65.1	1.55
4	0.278	75.9	1.08
5	0.256	85.8	0.99
6	0.144	91.4	0.56
7	0.100	95.3	0.39
8	0.089	98.7	0.34
9	0.033	100.0	0.13
10	0.000	100.0	0.00

Calling the top 10% reaches 2.8 times as many churners as random calls (the lift in the first decile), and the top 30% of scores contain about two-thirds of all churners. Gains and lift tables are how marketing and collections teams read models.

By target recall

Python

```
target = 0.80
i = np.where(rec ⩾ target)[0][-1]
print(f"to catch {target:.0%} of churners use threshold {pthr[min(i, len(pthr) - 1)]:.3f} →
precision {prec[i]:.3f}")
```

OUTPUT

```
to catch 80% of churners use threshold 0.245 → precision 0.465
```

Multiclass metrics

Python

```
from sklearn.datasets import load_wine
from sklearn.ensemble import RandomForestClassifier
w = load_wine(); Xa, Xb, ya, yb = train_test_split(w.data, w.target, test_size=0.4,
stratify=w.target, random_state=0)
pw = RandomForestClassifier(200, random_state=0).fit(Xa, ya).predict(Xb)
print(confusion_matrix(yb, pw))
for avg in ("macro", "weighted", "micro"): print(f"{avg:8s} F1 {f1_score(yb, pw,
average=avg):.3f}")
```

OUTPUT

```
[[23  1  0]
 [ 1 28  0]
 [ 0  0 19]]
macro    F1 0.975
weighted F1 0.972
micro    F1 0.972
```

Macro averages classes equally (rare classes count as much as common ones); **weighted** weights by class size; **micro** pools all decisions (equals accuracy for single-label problems).

Summary

KEY TAKEAWAYS

- Confusion matrix: TP, FP, FN, TN. Accuracy misleads on imbalanced data; compare with the majority baseline.
- Precision = how many flagged are right; recall = how many positives are caught; F1/F-beta balance them.
- ROC-AUC ranks quality independent of threshold; PR-AUC (average precision) is better for rare positives.
- Check log loss and calibration when probabilities drive decisions.
- Choose thresholds by cost, capacity (top-k, gains, lift) or target recall; use macro/weighted averages for multiclass.

Quick quiz

6 questions · tap an answer to check it

1. Fraud is 1% of transactions. A model that predicts "not fraud" for everything has accuracy...

A 1%

B 50%

C 99%, which is useless

D Undefined

Answer: C. That is why accuracy misleads on imbalanced data.

2. Recall answers which question?

A Of all actual positives, how many did we catch?

B Of all flagged cases, how many are right?

C How many negatives were correct?

D How confident is the model?

Answer: A. TP / (TP + FN).

3. A spam filter must not hide real emails. Which metric should be high?

A Precision for the spam class

B Recall for the spam class

C Specificity only for spam

D None

Answer: A. False positives (real mail marked spam) are costly.

4. ROC-AUC = 0.80 means...

A A random positive gets a higher score than a random negative 80% of the time

B 80% accuracy

C 80% recall

D 80% precision

Answer: A. It measures ranking quality.

5. Which curve is more informative when positives are very rare?

A ROC

B Precision-recall

C Calibration

D Learning curve

Answer: B. PR focuses on the positive class.

6. The retention team can call only 10% of customers. What should you report?

A Precision/recall and lift in the top 10% of scores

B Overall accuracy

C R^2

D Number of features

Answer: A. Evaluate where decisions are actually made.

EXERCISE

Practice

(1) Train a random forest on the same split and compare ROC-AUC, average precision and lift in the top decile with logistic regression. (2) Calibrate the forest with `CalibratedClassifierCV` and compare Brier scores. (3) Find the threshold that maximises F2.

Show solution (worked answer)

Python

```
from sklearn.calibration import CalibratedClassifierCV
rf = Pipeline([("pre", pre), ("rf", RandomForestClassifier(400, min_samples_leaf=10,
random_state=0))]).fit(Xtr, ytr)
pr = rf.predict_proba(Xte)[:, 1]; print(roc_auc_score(yte, pr),
average_precision_score(yte, pr))
top = np.argsort(-pr)[:len(pr) // 10]; print("lift", yte.to_numpy()[top].mean() /
yte.mean())
cal = Pipeline([("pre", pre), ("cal",
CalibratedClassifierCV(RandomForestClassifier(400, min_samples_leaf=10,
random_state=0), cv=5, method="isotonic"))]).fit(Xtr, ytr)
print(brier_score_loss(yte, pr), brier_score_loss(yte, cal.predict_proba(Xte)[:, 1]))
ts = np.linspace(.05, .9, 35); print(max(ts, key=lambda t: fbeta_score(yte, proba ⩾
t, beta=2)))
```

INTERVIEW CORNER

Questions you may be asked

Q: Precision vs recall: give an example where each matters more. Recall for disease screening or fraud (missing a case is costly); precision for spam filtering or expensive manual reviews (false alarms are costly).

Q: Why can a model with 95% accuracy be bad? With 95% negatives, predicting all negatives gives 95% accuracy and catches no positives; look at recall, precision, PR-AUC and the confusion matrix.

Q: How do you choose a classification threshold? From business costs and constraints: expected profit, team capacity (top-k), or a required recall/precision, validated on held-out data.

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